

A Report on:
“Tele-Clinic App”

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Prepared For Course:

Object Oriented Programming CS F213
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At:



BITS Pilani, Dubai Campus
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First Semester 2025-26

Date of Submission : 2nd December 2025

Table of Contents

1. Introduction
 - 1.1 Objectives
 - 1.2 Scope
2. System Analysis and Design
 - 2.1 Requirement Analysis
 - 2.1.1 Functional Requirements
 - 2.2 Use Case Diagram / Flowchart / UML Diagrams
 - 2.2.1 Use Case Summary
 - 2.2.2 System Flow (Text-Based Flowchart)
 - 2.3 Class Diagram
3. Implementation
 - 3.1 Explanation of Java Code Modules
 - 3.1.1 Frontend Modules
 - 3.1.2 Backend Modules (Telegram Bot)
 - 3.2 Main Java Concepts Used
 - 3.3 GUI
4. Conclusion
5. Future Works
6. References

Acknowledgements

We thank everyone who supported the successful completion of our Object-Oriented Programming (OOP) project.

We extend our heartfelt gratitude to our professors: Prof. Pranav Pawar, Prof. Sujala D. Shetty, Prof. Sapna Sadhwani, Prof. Kajal Kansal, Prof. Amanjot Kaur and Prof. Neena Shaji for their invaluable guidance, encouragement, and inspiration throughout this project.

Besides, we extend our appreciation to the other peers and team members for their collaboration and further contribution towards the project. Last but not least, to our institution, for resources and an environment in which to learn and grow and thus undertake and successfully complete such projects.

Abstract

This project aims to enhance the efficiency of clinic operations at BITS Pilani Dubai Campus. It focuses on reducing the response time required by clinic staff to provide immediate first aid to students. The system includes a dedicated emergency button that instantly sends an alert message from the nurse to the BPDC authorities. The students in case of an emergency can reach the nurse via the telegram bot which will alert the nurse and share the students exact geo-location. Additionally, the project features a centralized dashboard that stores and displays complete medical records and health histories for each student on campus. This integrated approach supports faster decision-making, improves accessibility to medical data, and ensures a more effective and timely emergency response.

1. Introduction

Tele-Clinic is designed to streamline emergency medical response and clinic operations at BITS Pilani Dubai Campus (BPDC). The system addresses critical delays in providing first aid and medical assistance to students, particularly during non-working hours and holidays. By integrating real-time communication tools, geolocation tracking, and centralized medical records, the project aims to significantly reduce response times and improve the quality of emergency care on campus.

Currently, the process of obtaining medical assistance at BPDC involves multiple manual steps and coordination among various parties. When a student requires urgent medical attention, the student must contact the in-house nurse and inform wardens. They will then verify insurance status and arrange transportation if necessary. This fragmented approach is time-consuming and inefficient, especially during emergencies when every minute counts.

The existing system lacks:

- **Real-time communication channels** between students and medical staff
- **Instant alert mechanisms** for emergencies
- **Location tracking** to identify students in distress quickly
- **Coordination tools** for efficient resource allocation and transport arrangement

These gaps create significant delays in emergency response, potentially compromising student health and safety. The absence of readily accessible medical histories further complicates decision-making during critical situations, as clinic staff and administrators cannot quickly assess pre-existing conditions, allergies, or insurance coverage.

The catalyst for this project came from a firsthand experience that highlighted the system's inadequacies. A student, Yash, suffered an ankle injury while playing volleyball at 11 PM on campus. Despite the urgency of the situation, he faced multiple obstacles: no wheelchair was immediately available, the nurse was difficult to reach outside regular hours, and his lack of university insurance added another layer of complexity. The process of coordinating with wardens and arranging transport took approximately two hours. An unacceptable delay for someone requiring immediate medical attention.

This incident underscored the urgent need for a systematic solution that can:

- **Reduce response times** from hours to minutes
- **Provide immediate access** to critical medical information
- **Streamline communication** between students, medical staff, and administrators
- **Enable informed decision-making** through readily available health records
- **Eliminate manual coordination bottlenecks** that delay emergency response

By developing this integrated emergency response system, we aim to ensure that no student at BPDC experiences unnecessary delays in receiving the medical care they need, regardless of the time of day or their insurance status.

1.1. Objectives:

1. **Reduce Emergency Response Time:** Minimize the time between a medical emergency occurrence and the delivery of first aid or medical assistance to under 15 minutes during all hours.
2. **Establish Real-Time Communication:** Implement an instant alert system that connects students, nurses, and campus authorities through a Telegram-based interface for immediate emergency notifications.
3. **Enable Location Tracking:** Integrate geolocation functionality to pinpoint the exact location of students requiring medical assistance, facilitating faster response.
4. **Centralize Medical Records:** Develop a comprehensive dashboard that stores and provides instant access to complete medical histories, insurance status, allergies, and emergency contact information for all students.
5. **Automate Emergency Workflows:** Streamline the coordination process among nurses, wardens, and administrators to eliminate manual bottlenecks in emergency response.
6. **Improve Medical Data Accessibility:** Ensure authorized personnel (nurses, administrators, and students themselves) can access relevant medical information anytime, anywhere.
7. **Enhance Decision-Making:** Provide clinic staff with immediate access to patient history to make informed decisions during emergencies.
8. **Support Non-Emergency Functions:** Enable appointment scheduling, prescription tracking, and routine health check-up management through the integrated system.

1.2 Scope:

1. Emergency Response System

- Telegram bot interface for students to report emergencies
- One-click emergency button for immediate alerts
- Automatic geolocation capture and sharing
- Push notifications to nurses and designated authorities

2. Centralized Dashboard

- App-based administrative interface for clinic staff and authorities
- Student medical records database including:
 - Personal information and emergency contacts
 - Medical history and pre-existing conditions

- Allergy information and medication records
- Insurance status and coverage details
- Past visit history and treatment records
- Role-based access control (Nurse, Student)
- Search and filter functionality for quick record retrieval

3. Communication Module

- Multi-channel alert system (Telegram)
- Automated notification to wardens and transport services
- Group communication for coordinated response
- Status confirmation and acknowledgment system

4. User Management

- Student profile creation and management
- Staff and authority account management
- Permission and role assignment
- Authentication and security protocols

2. System Analysis and Design

[\(This is the demo link to our project.\)](#)

2.1 Requirement Analysis

2.1.1 Functional Requirements

1. User Login System

- Students and administrators must be able to log into the GUI using credentials stored in credentials.csv.

2. Medical Registration Form

- Students must be able to submit medical information via the GUI.
- Data is saved in medical_registrations.csv.

3. Student Dashboard

- View submitted medical information.
- Access prescriptions from prescriptions.txt.

4. Admin Dashboard

- View all student medical submissions.
- Monitor emergency logs.
- Manage student records.

5. Emergency Alert System

- Students/admins can trigger an emergency alert.
- The alert is stored in emergency_logs.txt.

6. **Telegram Bot Notification System (Backend)**
 - The backend monitors messages.txt and emergency_logs.txt.
 - It maps student IDs to guardian IDs using medical_registrations.csv.
 - Sends Telegram notifications to guardians.
7. **File-based Communication Between Frontend & Backend**
 - The frontend writes logs.
 - The backend reads logs and triggers actions.

2.1.2 Non-Functional Requirements

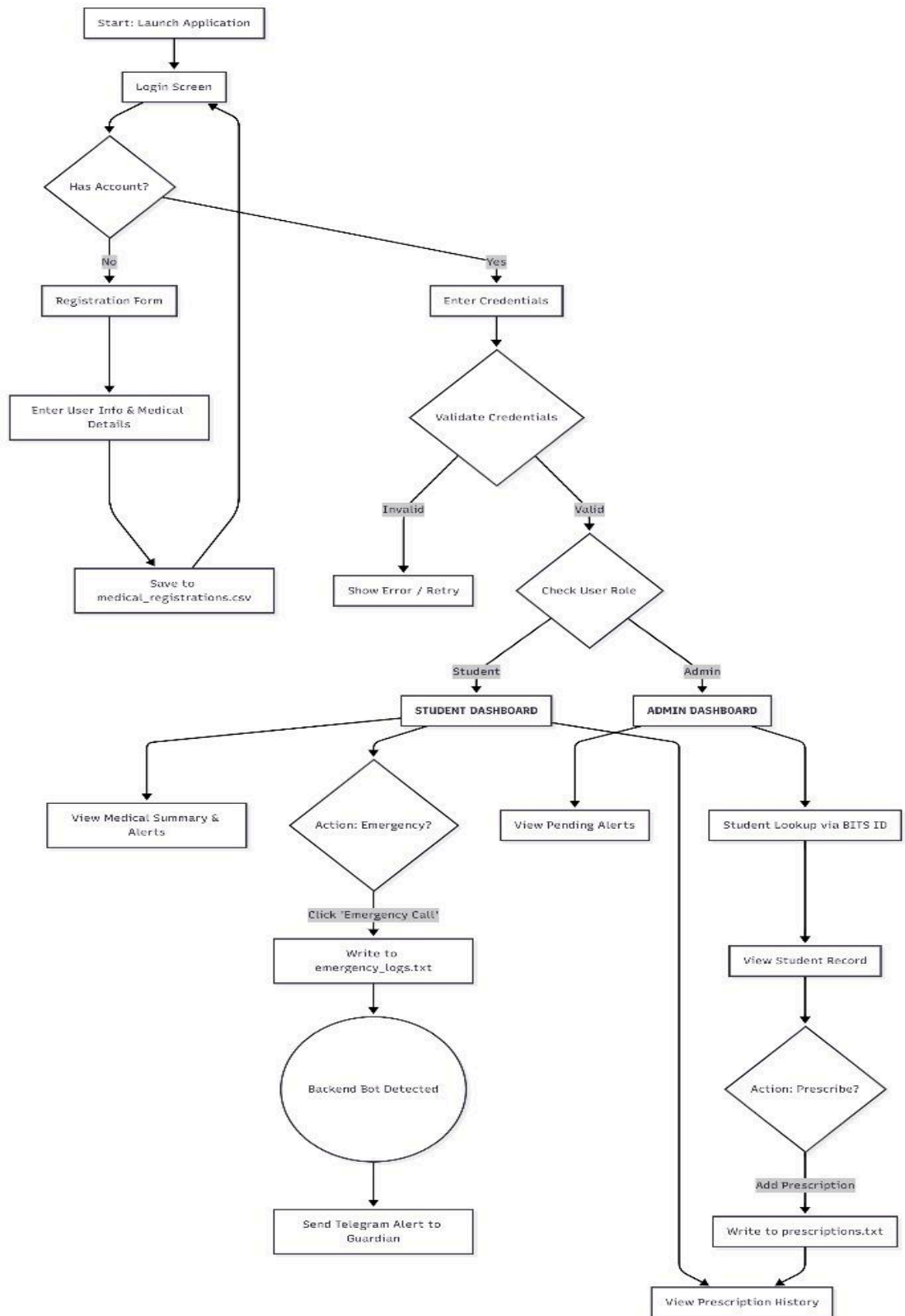
1. **Usability**
 - GUI should be simple, intuitive, and accessible.
2. **Portability**
 - Runs on Java 8+ and Maven 3.9+ without OS dependencies.
3. **Performance**
 - Backend must continuously listen to logs with minimal delay.
4. **Reliability**
 - Accurate reading/writing of CSV and text files.
5. **Maintainability**
 - Modular code organization (app, core, util, and backend modules).
6. **Security**
 - Credentials stored in CSV—basic confidentiality.
 - Restricted access to admin dashboard.

2.2 Use Case Diagram / Flowchart / UML Diagrams

2.2.1 Use Case Summary

Actor	Use Cases
Student	Login, Submit Medical Form, View Dashboard, Trigger Emergency Alert
Admin	Login, View All Registrations, Monitor Emergencies
Telegram Bot (System)	Listen to Logs, Send Notifications

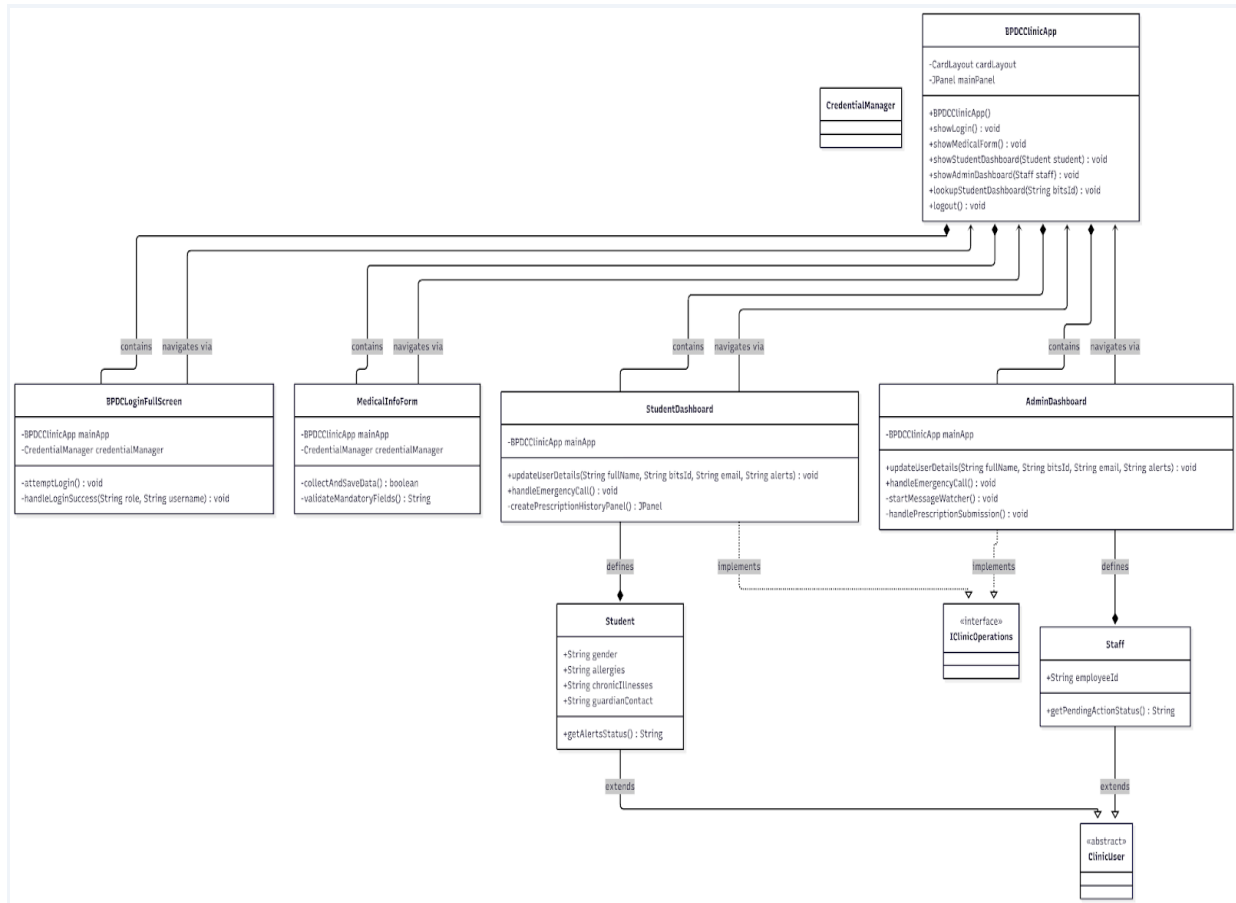
2.2.2 System Flow (Text-Based Flowchart)



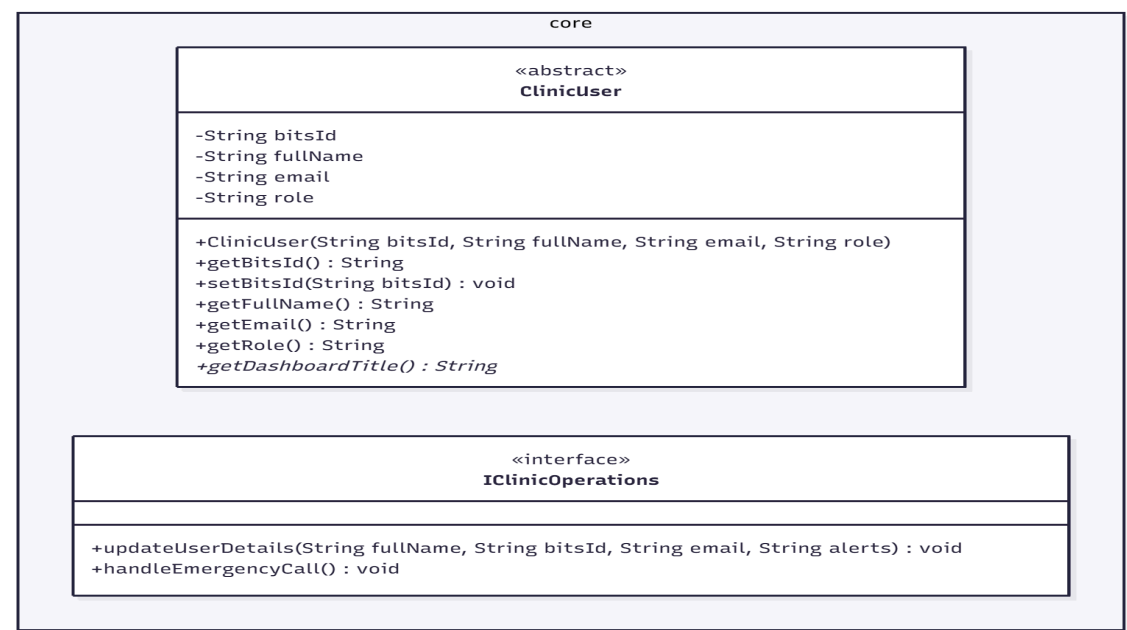
2.3 Class Diagram

Frontend Packages

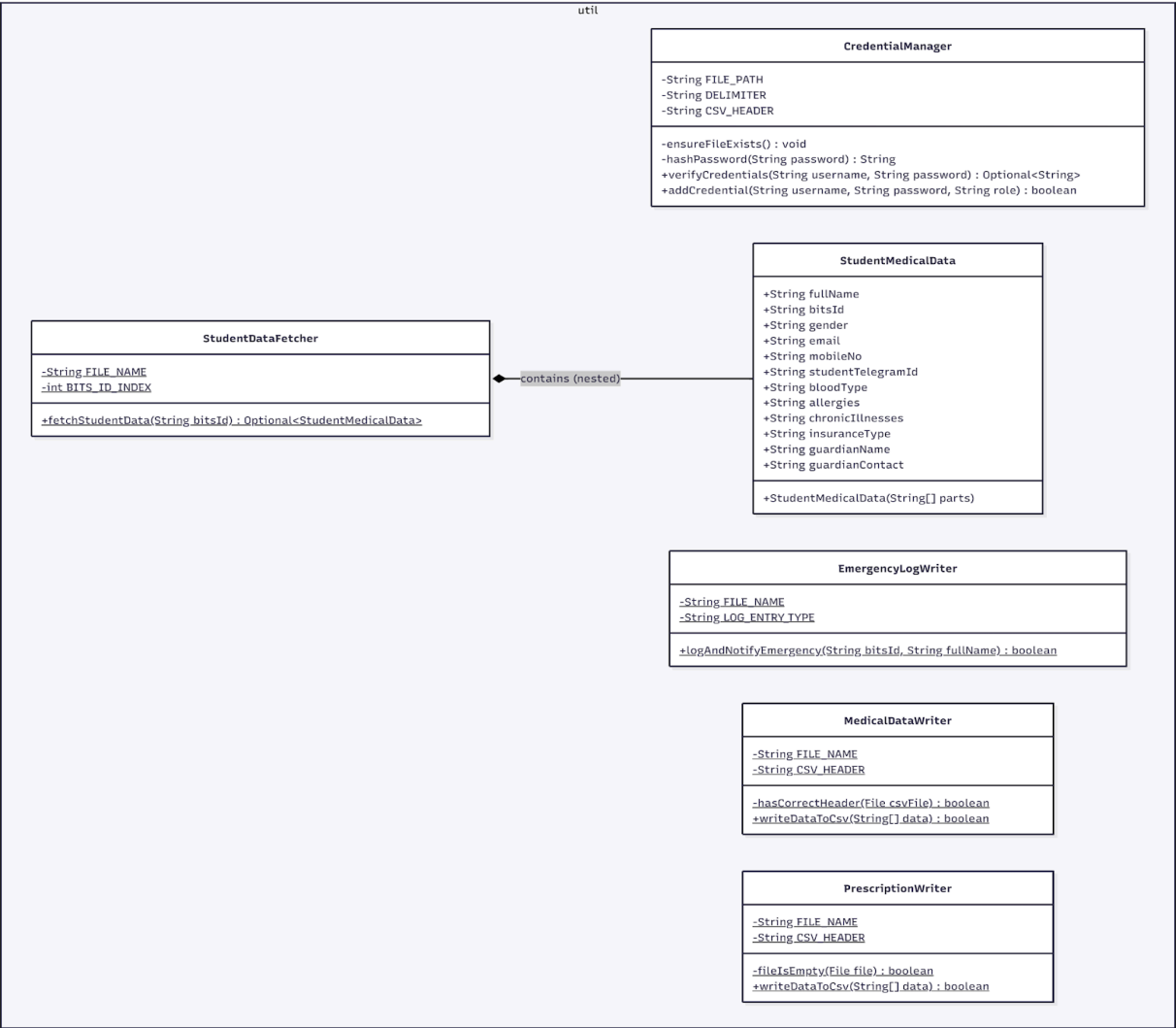
Package: app



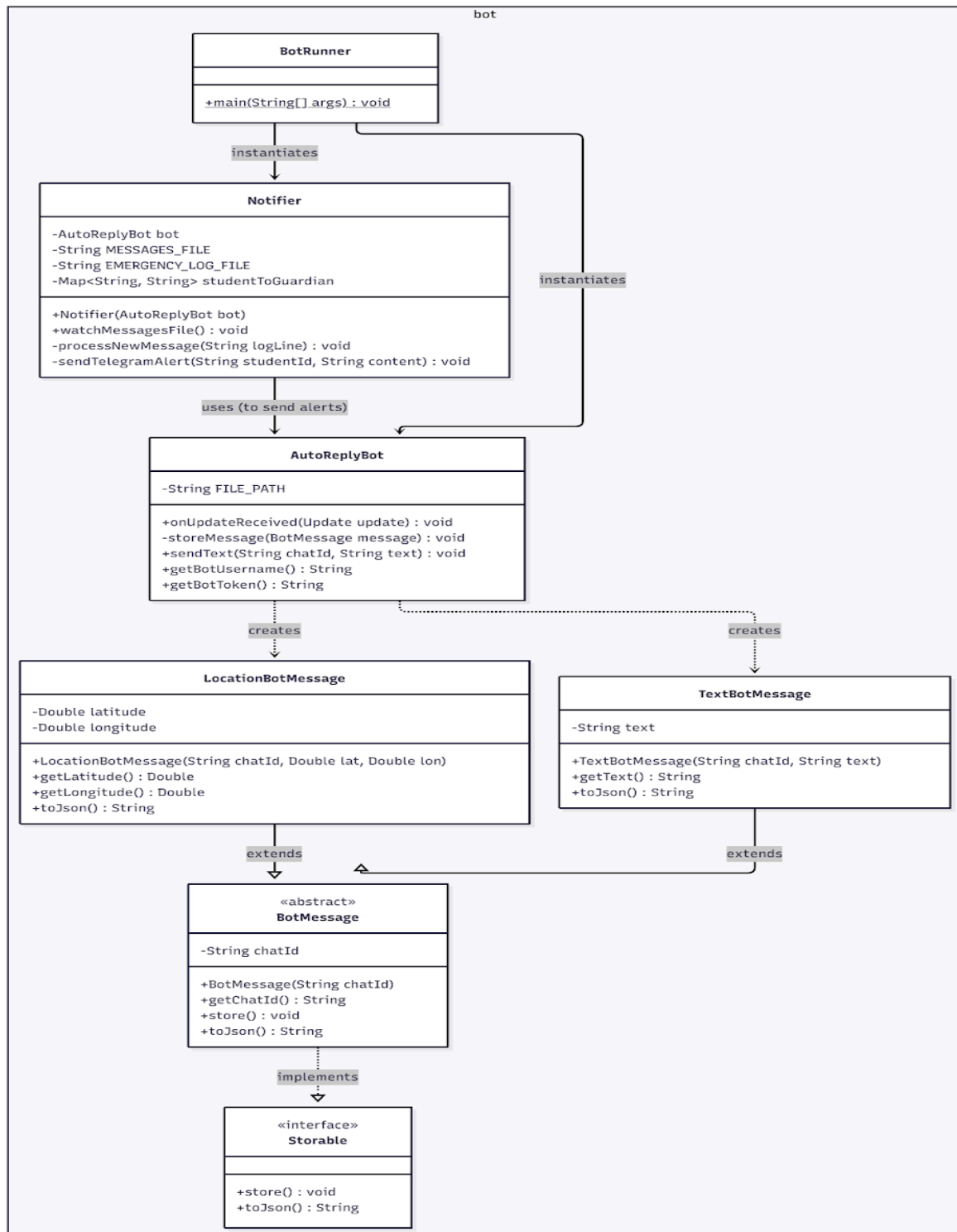
Package: core



Package: util



Backend (Telegram Bot)



3. Implementation

Source Code: https://github.com/himeshsoni704/TELE_CLINIC

Demo: ([This is the demo link to our project.](#))

3.1 Explanation of Java Code Modules

3.1.1 Frontend Modules

1. app Package

The app package manages the overall GUI flow of the Tele_Clinic application. It controls how users navigate from the login screen to their respective dashboards and form interfaces.

2. core Package

The core package contains the main data models of the system focused on representing system entities and managing their behavior.

- **User** (base class)
- **Student** and **Admin** (specialized roles that extend User)

3. util Package

The util package handles all file operations and reusable helper functions, keeping the code organized. It helps in performing the following functions and ensures the frontend can reliably access and store data.

- Reading CSV files
- Writing medical records or credentials
- Loading and saving text-based logs

4. Modules

javax.swing

Used for creating windows, buttons, labels, and form components.(JFrame, JPanel, JButton, JLabel, JTextField, JTable)

java.util

Used for handling lists, simple data processing, storing user data, managing collections, processing form inputs, parsing CSV lines, splitting strings, managing

lists, and utility operations

java.io

Used to input and output data, used for reading and writing in files as well (FileReader, FileWriter, BufferedReader, etc.)

Java.awt

Work as Layout managers (BorderLayout, GridLayout), colors, fonts, and event handling, provides structure, visual styling for the interface, layout managers, fonts, colors, and basic GUI styling

File Name	Application
app/BPDCClinicApp.java	Main Application Manager: Initializes the GUI, sets up navigation (Login, Dashboards, Registration), and manages the active user session.
app/BPDCLoginFullScreen.java	Login interface: Handles user authentication against credentials.csv and routes the user to the correct dashboard (Student or Admin/Nurse).
app/MedicalInfoForm.java	Registration Form: GUI to collect a new user's personal and medical data, registers their login credentials, and saves their record to medical_registrations.csv.
app/StudentDashboard.java	Student Interface: Displays the student's medical summary, loads prescription history from prescriptions.txt, and allows the user to trigger the emergency alert.

app/AdminDashboard.java	Staff Interface: Allows nurses/admins to look up student records, create and save new prescriptions to prescriptions.txt, and view real-time notifications from the backend bot.
util/StudentDataFetcher.java	Access specific student's records from medical_registrations.csv for display on the dashboards.
util/EmergencyLogWriter.java	Writes an entry to the emergency_logs.txt file when the Student Dashboard's emergency button is pressed.
core/ClinicUser.java	Serves as the base class for all user types (Student, Staff). It defines common attributes (bitsId, fullName, email, role) and an abstract method (getDashboardTitle()) that subclasses must implement.
core/IClinicOperations.java	Defines a mandatory contract for all dashboard panels (like StudentDashboard and AdminDashboard). It ensures they implement key operations such as updateUserDetails() and handleEmergencyCall().

3.1.2 Backend Modules (Telegram Bot)

The backend continuously monitors the frontend's output files and sends automated notifications to guardians. It operates independently from the GUI but communicates with it through shared text and CSV files.

1. Files Monitored

- messages.txt
Contains general notifications or student messages written from the GUI.
- emergency_logs.txt
Stores emergency alerts triggered by students or admins from the frontend.

The bot monitors the given files, hereby reacting to emergencies instantly.

2. Files Accessed

To send accurate guardian messages, the bot reads “medical_registrations.csv”, which is used to map student IDs to guardian contact information. The CSV processing is handled using Java file I/O and string utilities.

3. Technologies & Libraries Used

- Java File Watchers
Implemented using `java.nio.file.WatchService` to detect real-time changes in log files.
- Telegram API Libraries
Maven dependencies are used to connect the bot to the Telegram Bot API, enabling:
 - reading bot updates
 - sending messages
 - managing chat sessions

4. Notifications Sent by the Bot

The backend bot automatically sends:

- Emergency Alerts
When a new entry appears in `emergency_logs.txt`, the bot instantly notifies the relevant guardian on Telegram.
- Message Notifications
Any general messages in `messages.txt` are forwarded as appropriate Telegram alerts.

Files	Application
AutoReplyBot.java	Telegram Bot Core: Handles all inbound Telegram messages, commands (/start), location data, and sends automated replies to users.

Notifier.java	Alert System: Monitors log files (messages.txt, emergency_logs.txt) and, upon detecting an urgent entry, sends a Telegram alert to the relevant student's guardian.
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3.2 Main Java Concepts Used

1. Inheritance

- The Abstract ClinicUser class serves as the base for role-specific classes, ensuring common fields like bitsId and email are defined only once. Both Student and AdminDashboard.Staff extend this base class to inherit foundational user properties.

2. Abstraction

- The Abstract ClinicUser class defines the essential methods like viewDashboard() as abstract, forcing concrete implementations in subclasses. The FileHandler utility class hides the complex details of java.io (like opening BufferedReaders and handling exceptions) behind simple, clean static methods.

3. Polymorphism

- The system treats all authenticated users as the general ClinicUser type. When a core method like getDashboardTitle() is called, the runtime determines whether to execute the Student version ("BPDC Clinic Student Portal") or the Staff version ("BPDC Clinic Nurse Portal").

4. Encapsulation

- All data attributes within domain classes (e.g., in ClinicUser or StudentDataFetcher.StudentMedicalData) are declared as private. Access and modification are strictly controlled via public getter methods and controlled setter methods (where available) to ensure data validation.

5. Interfaces

- The IClinicOperations interface dictates mandatory methods like handleEmergencyCall() and updateUserDetails() that must be implemented by both the StudentDashboard and AdminDashboard.
- The Storable interface ensures all backend message classes (TextBotMessage, LocationBotMessage) provide the required toJson() method for persistent logging.

6. Modular Architecture / Separation of Concerns

- The project is divided into two major processes (Frontend GUI and Backend Telegram Bot) that operate independently. Within the Frontend, functionality is logically segregated into three packages: app (Presentation/GUI), core (Business Logic/Domain Models), and util (Infrastructure/File I/O).

7. File Handling

- The Frontend writes emergency events to emergency_logs.txt. The Backend utilizes advanced file system monitoring (WatchService and RandomAccessFile) to detect new entries in the log files instantly, acting as a lightweight, reliable asynchronous messaging system.

8. Exception Handling

- try-with-resources blocks are used extensively in file I/O classes (e.g., MedicalDataWriter.java) to guarantee that resources like FileWriter are closed, even if IOExceptions occur. Separate catch blocks handle specific errors like API failures or file read issues gracefully.

3.3 GUI

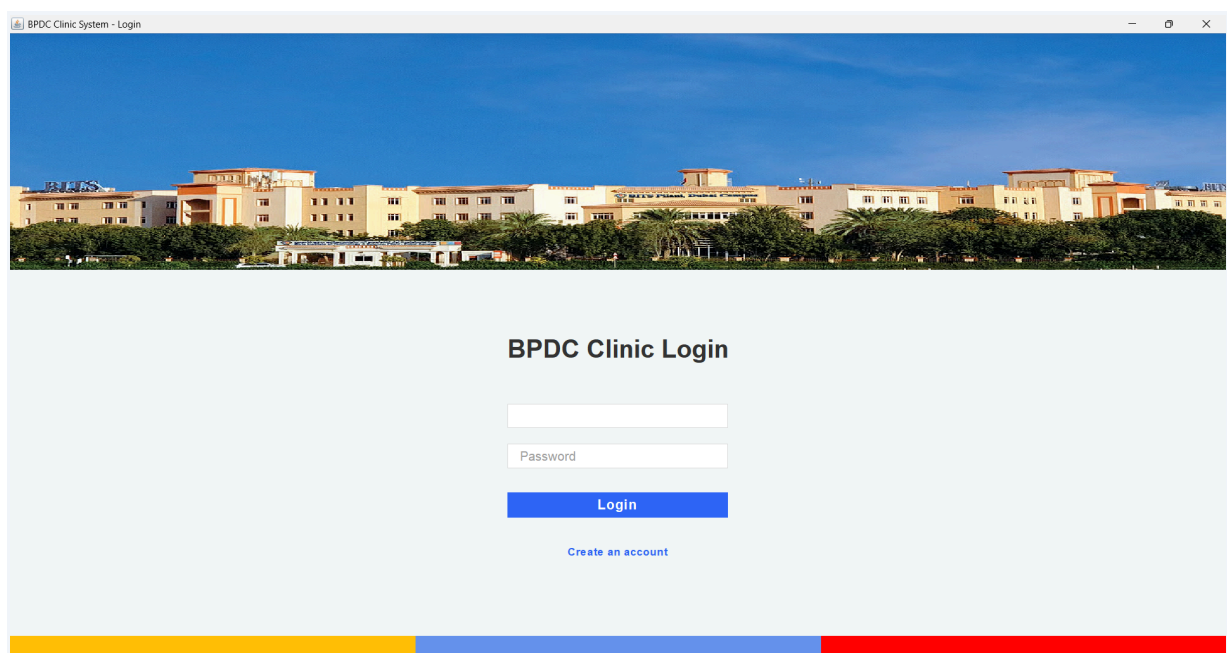


Fig.1 Login Page

BPDC Clinic System - Medical Registration

USER INFORMATION

Full Name

BITS ID

Set Account Password

Gender (Select)

BITS Email

+971 Mobile No. +971 Telegram ID

D.O.B : DD MM YYYY

Emirates ID

Guardian's Name

+971 Guardian's contact no.

Guardian's email address

MEDICAL INFORMATION

Blood Type

Allergies (N/A if none)

Chronic Illnesses/Diseases (N/A if none)

Current medication

Past medication/Treatments

Insurance Type

← CREATE

Fig.2 Create an Account(Medical Form)

BPDC Clinic Student Portal

Welcome, Vignesh Radhakrishnan

f20240254@dubai.bits-pilani.ac.in

Student ID: 2024A7PS0255U

EMERGENCY CALL LOGOUT

Medical Summary

Gender: Male

Blood Type: AB+ve

Insurance: Institute Insurance

Guardian Contact

Name: Radhakrishnan

Contact No: 5625334180

Last Prescription

2025-11-30 13:24:00

Paracetamol

View Full History →

Nearby Medical Facilities

Fakeeh University Hospital

Dubai Silicon Oasis (Hospital)

Aster Clinic

Dubai Silicon Oasis (Clinic)

...View Full List

Medical Alerts Detail (Allergies & Chronic Illnesses)

Allergies: Flu

Chronic Illnesses/Diseases: Diabetes

Fig.3 Student Dashboard(Login using Credentials)

BPDC Clinic Nurse Portal

Welcome, Clinic Staff

ID: ADMIN_001

Dashboard Prescription Student Lookup

Quick Lookup: Enter BITS ID SEARCH

BITS Email admin_001@bpdclinic.com

Medical Alerts 3 Pending appointments to review.

Active Clinic BPDC Main Clinic

Fig.4 Admin Dashboard

BPDC Clinic Nurse Portal

Welcome, Clinic Staff

ID: ADMIN_001

Dashboard Prescription Student Lookup

Add Prescription

Student BITS ID: 2024A7PS0255U

Medication / Diagnosis: Paracetamol

Additional Instructions: Thrice a Day

SAVE PRESCRIPTION

Fig.5 Admin Dashboard(Prescription)

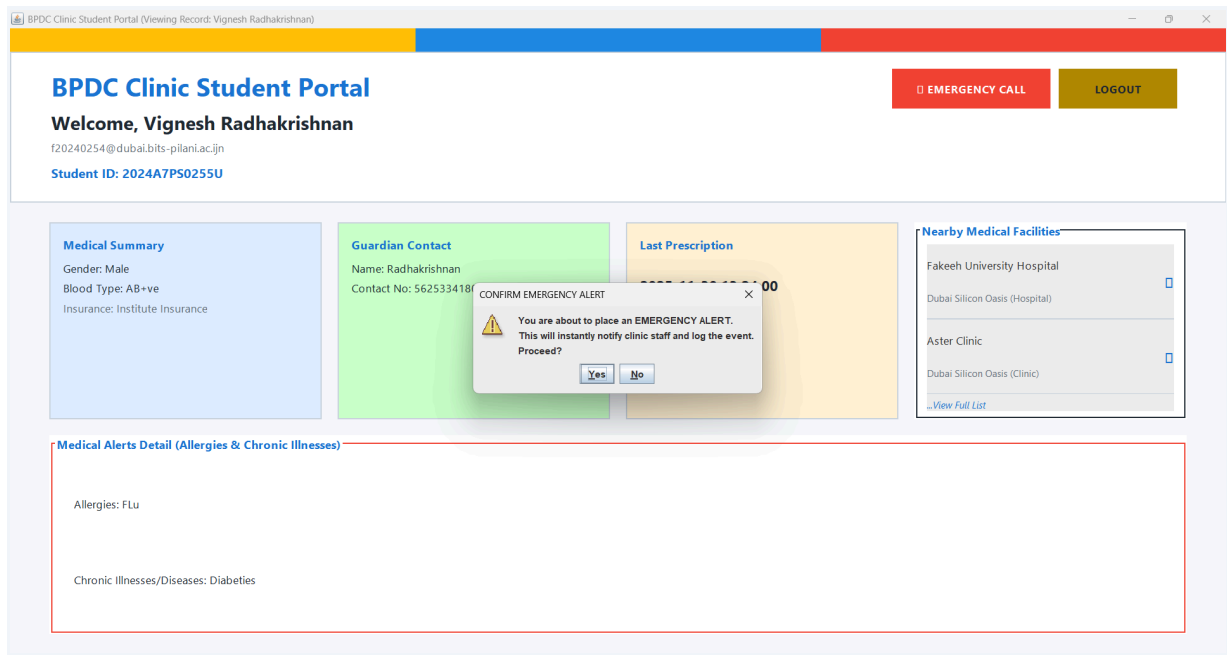


Fig.6 Student Dashboard(Emergency)

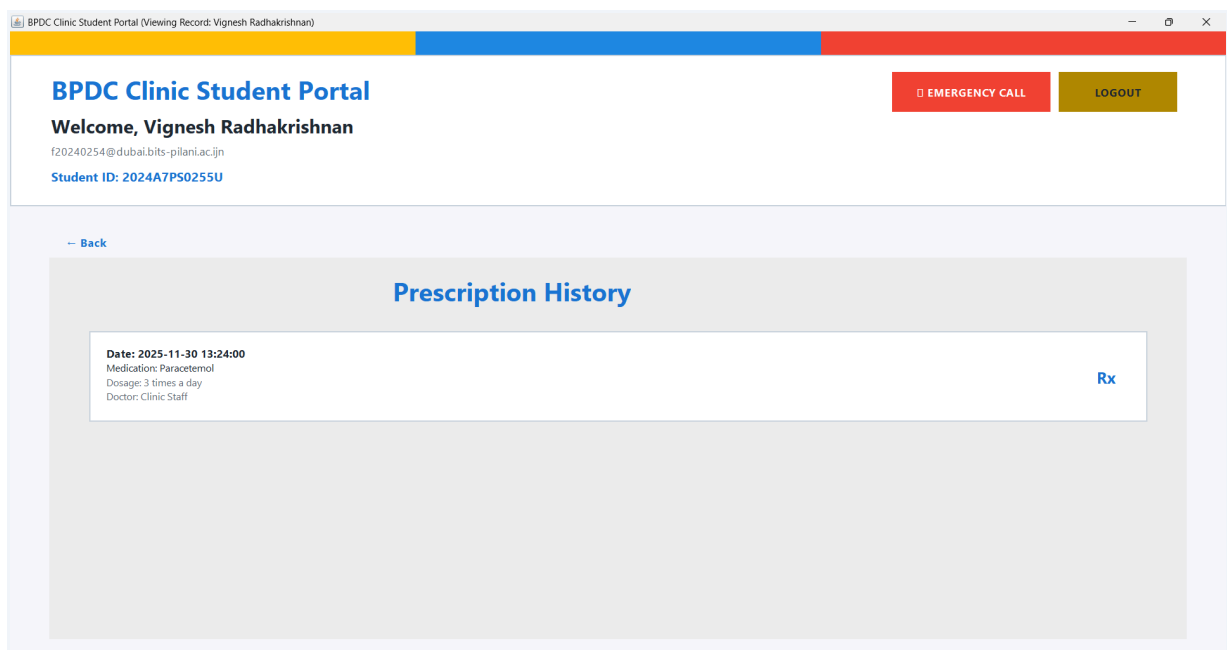
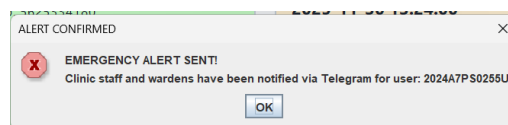


Fig.7 Student Dashboard(Prescription History)

4. Conclusion

Tele-Clinic successfully addresses critical gaps in emergency medical response at BITS Pilani Dubai Campus by transforming response times from hours to minutes. Through a Telegram-based alert system, real-time geolocation tracking, and centralized medical records dashboard, the project has created an efficient, automated solution that prioritizes student safety.

The system benefits everyone concerned, students receive immediate medical assistance regardless of time or location, medical staff make important decisions with instant access to patient histories, and administrators get streamlined coordination with incident tracking. The implementation effectively demonstrates object-oriented programming principles to create a secure and balanced solution.

Beyond technical accomplishments, this project represents a fundamental shift from manual processes to technology-driven emergency response. It creates a replicable model for other educational institutions facing similar challenges.

What began as a response to a friend's delayed medical care has evolved into a comprehensive solution with real potential to transform campus healthcare delivery. This project demonstrates how thoughtful application of technology solves real-world problems and creates meaningful impact, ensuring that no student experiences unnecessary delays in receiving critical medical attention.

5. Future Works

1. Facilitate Resource Management: Track availability of medical resources (wheelchairs, ambulances, first-aid supplies) and enable efficient allocation during emergencies.
2. Future scope can be actually making this live by hosting it on the web.
3. Connect with smartwatches and fitness trackers (Apple Watch, Fitbit, Samsung Galaxy Watch)
4. Follow-up appointment scheduling and reminders.
5. Virtual queue management for clinic visits.
6. SMS fallback when internet connectivity is unavailable
7. Automated medication reminders and adherence tracking.
8. Mental health assessment tools and counseling appointment management
9. Integration with external hospital records for comprehensive health history.

References

- [1] Z. Hasanov, "How to create Telegram Bot in Java [Tutorial]," *YouTube*, Feb. 2018.
- [2] T. S. Funny, "Java GUI CRUD Tutorial: Step-by-Step Guide for Beginners," *YouTube*, Dec. 2024.
- [3] R. Bermudez, "Telegram Bots," *GitHub Repository*, 2024.
- [5] C. S. Horstmann, *Object-Oriented Design & Patterns*, 2nd ed. Hoboken, NJ, USA: John Wiley & Sons, 2005.
- [6] H. Schildt and D. Coward, *Java: The Complete Reference*, 13th ed. New York, NY, USA: McGraw-Hill, 2024.
- [7] M. J. Chen, T. Y. Liu, and C. H. Wu, "Real-Time Location-Based Emergency Medical Service System for Campus Healthcare," *IEEE Access*, vol. 8, pp. 145632-145644, 2020.
- [8] R. Patel, S. Kumar, and A. Mishra, "Telegram Bot Framework for Healthcare Alert Systems: Architecture and Implementation," in *Proc. IEEE Int. Conf. Intelligent Systems and Green Technology (ICISGT)*, Visakhapatnam, India, 2019, pp. 281-286.
- [9] J. W. Cooper, *Java Design Patterns: A Tutorial*. Boston, MA, USA: Addison-Wesley, 2000.